

Spring REST using Spring Boot 3

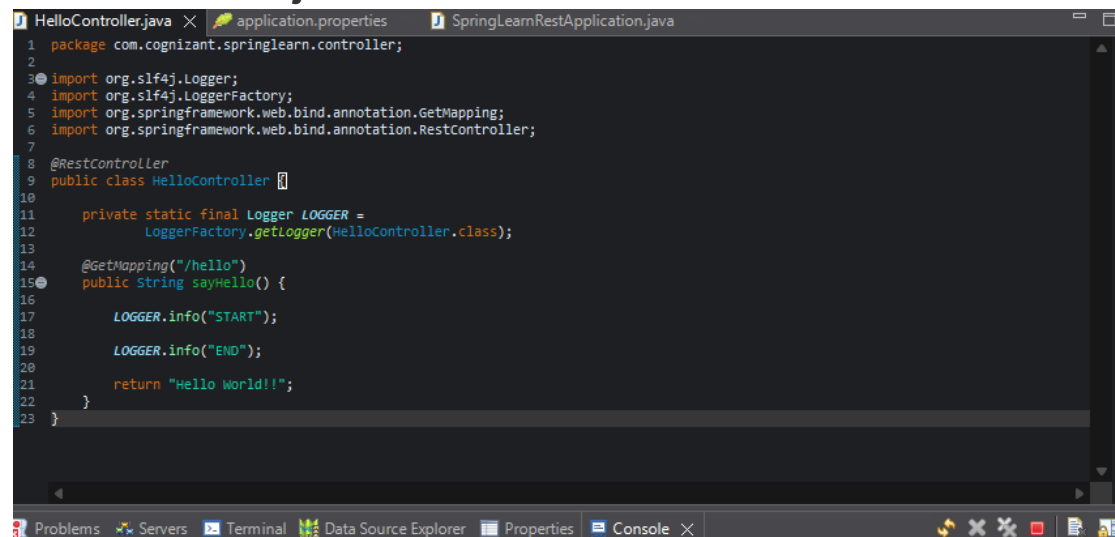
2.Spring-Rest-hands-on

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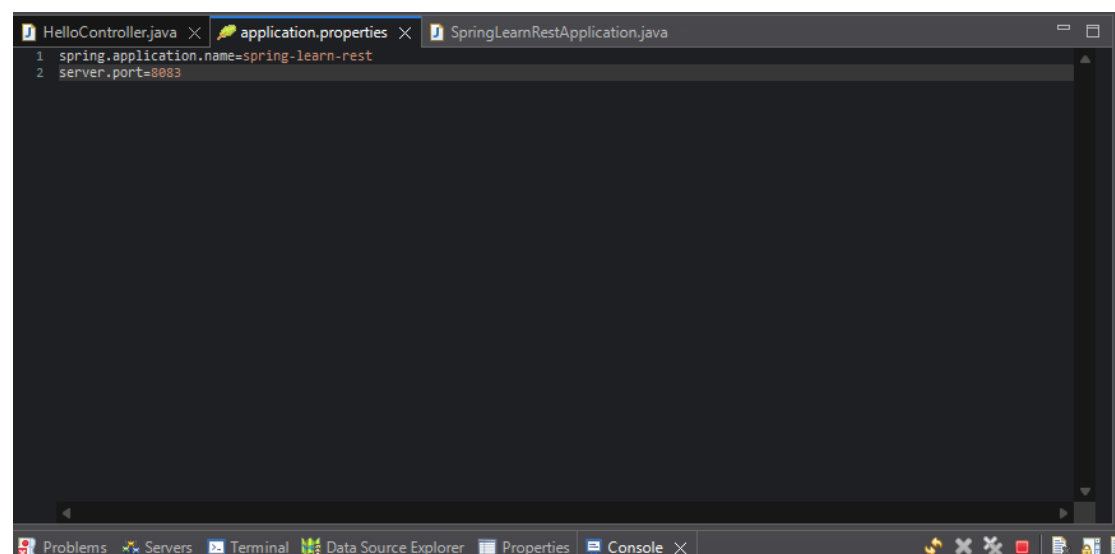
Hello World RESTful Web Service:

HelloController.java:



```
1 package com.cognizant.springlearn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.web.bind.annotation.GetMapping;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class HelloController {
10
11     private static final Logger LOGGER =
12         LoggerFactory.getLogger(HelloController.class);
13
14     @GetMapping("/hello")
15     public String sayHello() {
16
17         LOGGER.info("START");
18
19         LOGGER.info("END");
20
21         return "Hello World!!";
22     }
23 }
```

Application.properties:



```
1 spring.application.name=spring-learn-rest
2 server.port=8083
```

SpringLearnRestApplication.java:

```
1 package com.cognizant;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class SpringLearnRestApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(SpringLearnRestApplication.class, args);
11     }
12
13 }
14
```

Output:

```
WEEK-2 TASKS ECLIPSE - spring-learn-rest/src/main/java/com/cognizant/SpringLearnRestApplication.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer
LibraryManagementExercise2
LibraryManagementExercise4
orm-learn [boot] [devtools]
spring-learn [boot] [devtools]
spring-learn-handson2 [boot] [devtools]
spring-learn-rest [boot] [devtools]
src/main/java
com.cognizant
SpringLearnRestApplication.java
com.cognizant.springlearn.controller
HelloController.java
static
templates
application.properties
src/test/java
JRE System Library [JavaSE-17]
Maven Dependencies
target
HELP.md
mvnw
mvnw.cmd
pom.xml

SpringLearnRestApplication.java
application.properties
SpringLearnRestApplication.java

package com.cognizant;
import org.springframework.boot.SpringApplication;
@SpringBootApplication
public class SpringLearnRestApplication {
    public static void main(String[] args) {
        SpringApplication.run(SpringLearnRestApplication.class, args);
    }
}

Outline
com.cognizant
SpringLearnRestApplication
main(String[]) void

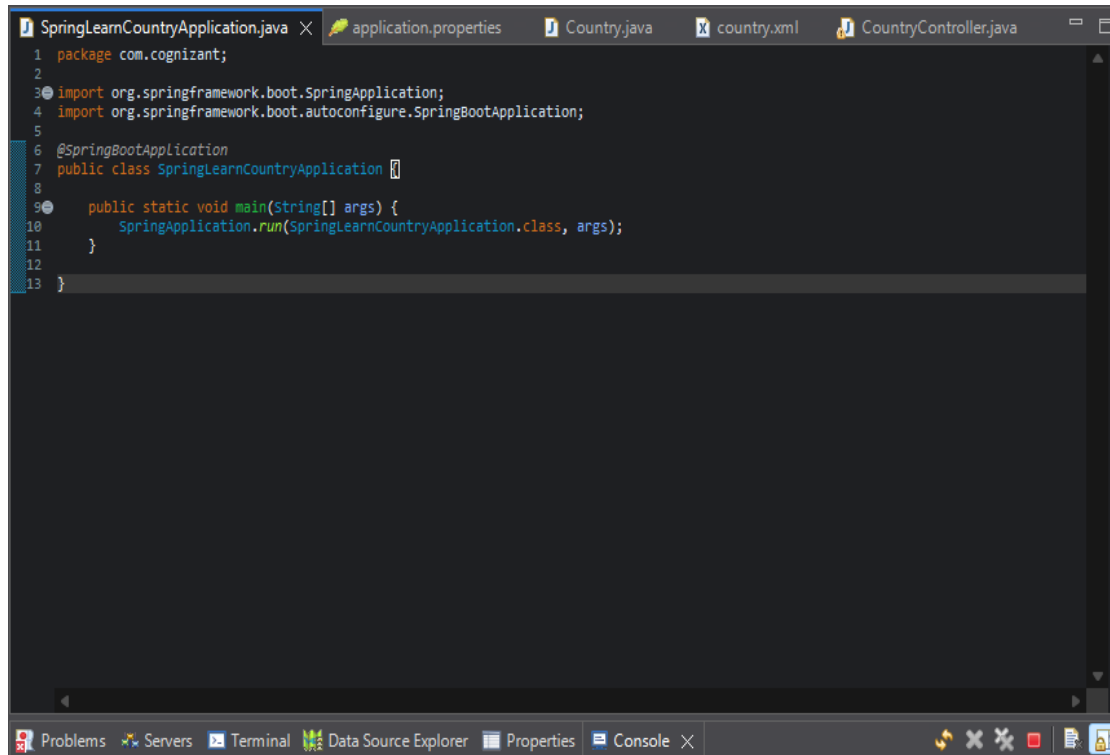
Spring
Spring Boot 1.0 (v4.1.0)

2026-07-02T14:32:44.343+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] c.cognizant.SpringLearnRestApplication : Starting SpringLearnRestApplication using Java 21.
2026-07-02T14:32:44.352+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] c.cognizant.SpringLearnRestApplication : No active profile set, falling back to 1 default p
2026-07-02T14:32:44.383+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] o.devtools.property.defaults.PostProcessor : DevTools property defaults active Set 'spring.dev
2026-07-02T14:32:44.384+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] o.devtools.property.defaults.PostProcessor : For additional web related logging consider settin
2026-07-02T14:32:46.416+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-02T14:32:46.473+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-02T14:32:46.474+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-02T14:32:46.592+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization complet
2026-07-02T14:32:47.596+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context pa
2026-07-02T14:32:47.608+05:30 INFO 2264 --- [spring-learn-rest] [ restartedMain] c.cognizant.SpringLearnRestApplication : Started SpringLearnRestApplication in 4.644 second
2026-07-02T14:33:51.796+05:30 INFO 2264 --- [spring-learn-rest] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Spring dispatcherServlet 'dispatcher
2026-07-02T14:33:51.796+05:30 INFO 2264 --- [spring-learn-rest] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-02T14:33:51.860+05:30 INFO 2264 --- [spring-learn-rest] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
2026-07-02T14:33:51.860+05:30 INFO 2264 --- [spring-learn-rest] [nio-8083-exec-1] c.c.s.controller.HelloController : START
2026-07-02T14:33:51.860+05:30 INFO 2264 --- [spring-learn-rest] [nio-8083-exec-1] c.c.s.controller.HelloController : END
```

```
localhost:8083/hello
Hello World!!
```

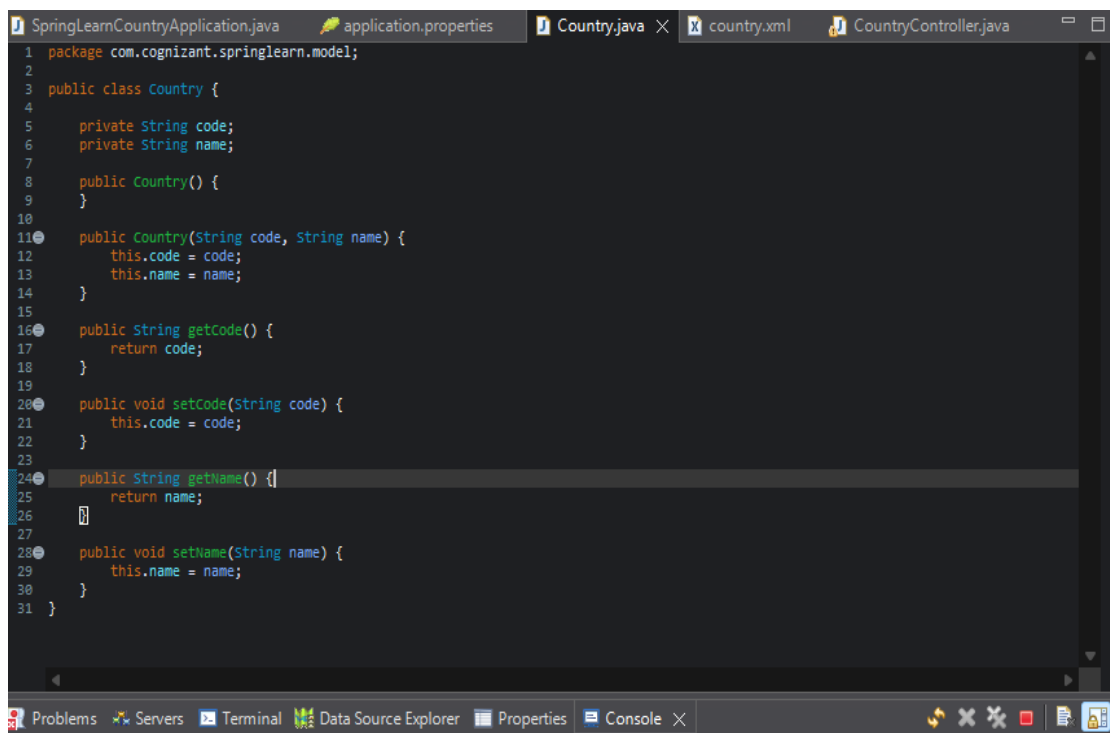
REST - Country Web Service:

SpringLearnCountryApplication.java:



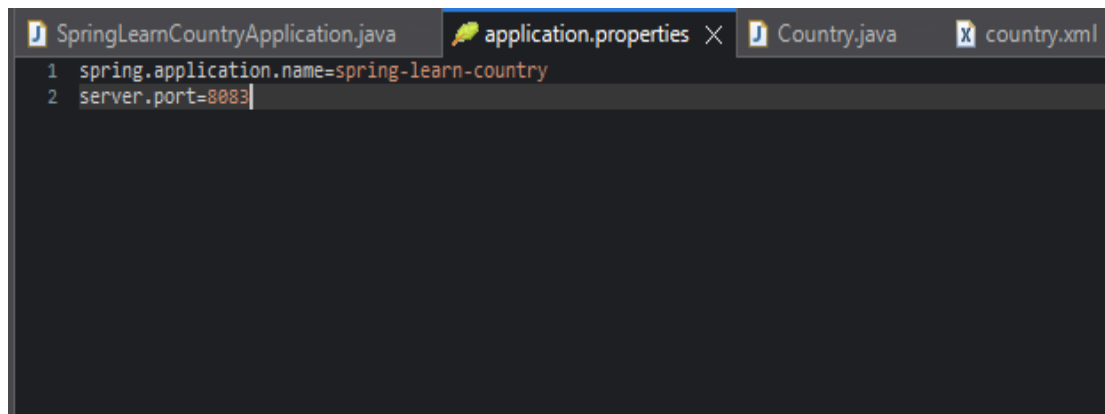
```
1 package com.cognizant;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class SpringLearnCountryApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(SpringLearnCountryApplication.class, args);
11     }
12
13 }
```

Coutry.java:



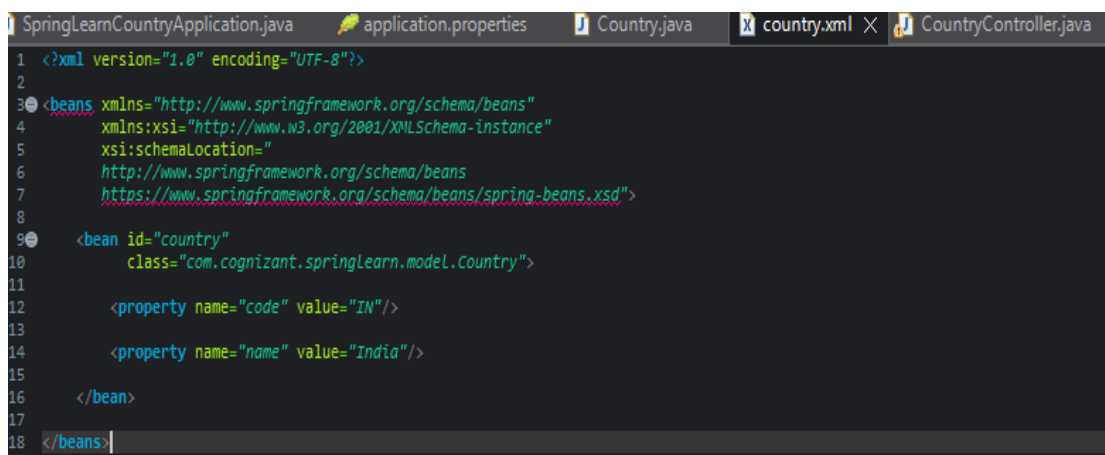
```
1 package com.cognizant.springlearn.model;
2
3 public class Country {
4
5     private String code;
6     private String name;
7
8     public Country() {
9     }
10
11     public Country(String code, String name) {
12         this.code = code;
13         this.name = name;
14     }
15
16     public String getCode() {
17         return code;
18     }
19
20     public void setCode(String code) {
21         this.code = code;
22     }
23
24     public String getName() {
25         return name;
26     }
27
28     public void setName(String name) {
29         this.name = name;
30     }
31 }
```

Application.properties:

A screenshot of an IDE window showing the 'application.properties' file. The tabs at the top are 'SpringLearnCountryApplication.java', 'application.properties', 'Country.java', and 'country.xml'. The code in the editor is:

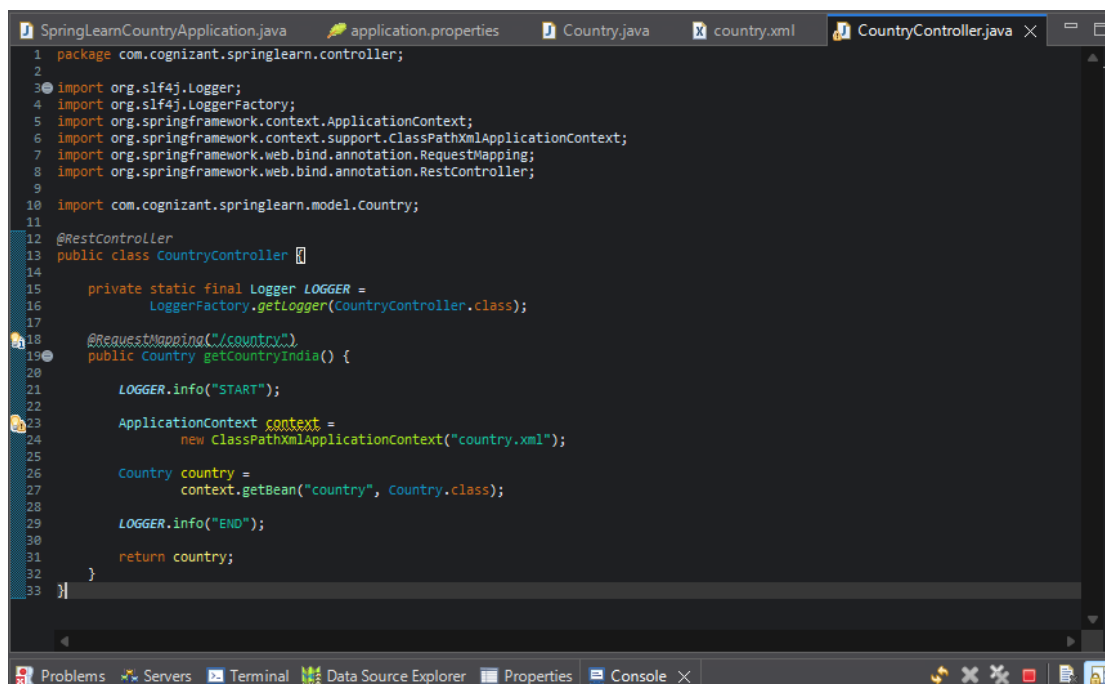
```
1 spring.application.name=spring-learn-country
2 server.port=8083
```

Country.xml:

A screenshot of an IDE window showing the 'country.xml' file. The tabs at the top are 'SpringLearnCountryApplication.java', 'application.properties', 'Country.java', 'country.xml', and 'CountryController.java'. The code in the editor is:

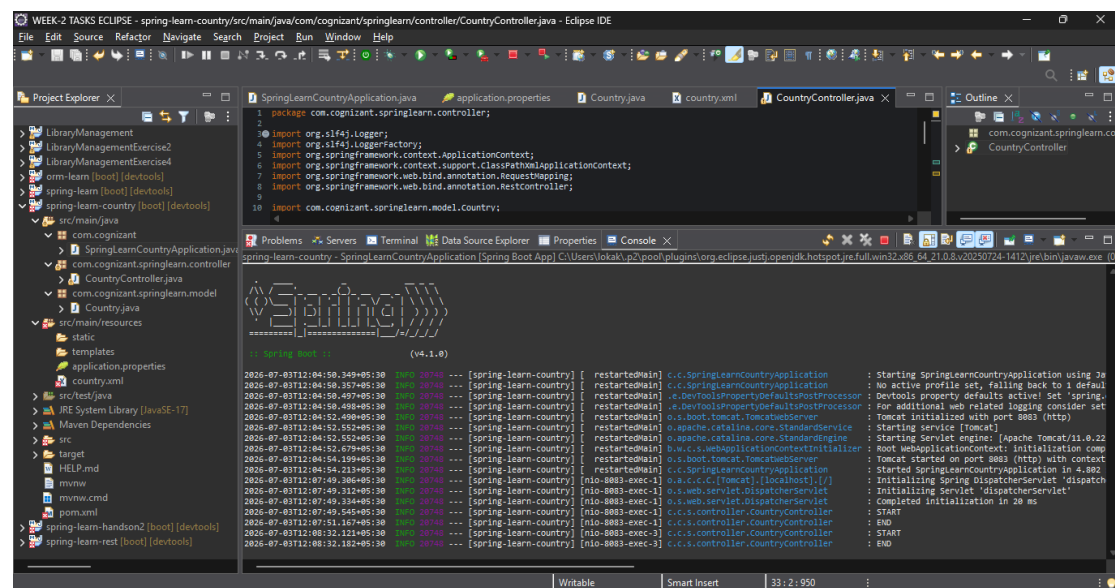
```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9   <bean id="country"
10        class="com.cognizant.springlearn.model.Country">
11
12     <property name="code" value="IN"/>
13
14     <property name="name" value="India"/>
15
16   </bean>
17
18 </beans>
```

CountryController.java:

A screenshot of an IDE window showing the 'CountryController.java' file. The tabs at the top are 'SpringLearnCountryApplication.java', 'application.properties', 'Country.java', 'country.xml', and 'CountryController.java'. The code in the editor is:

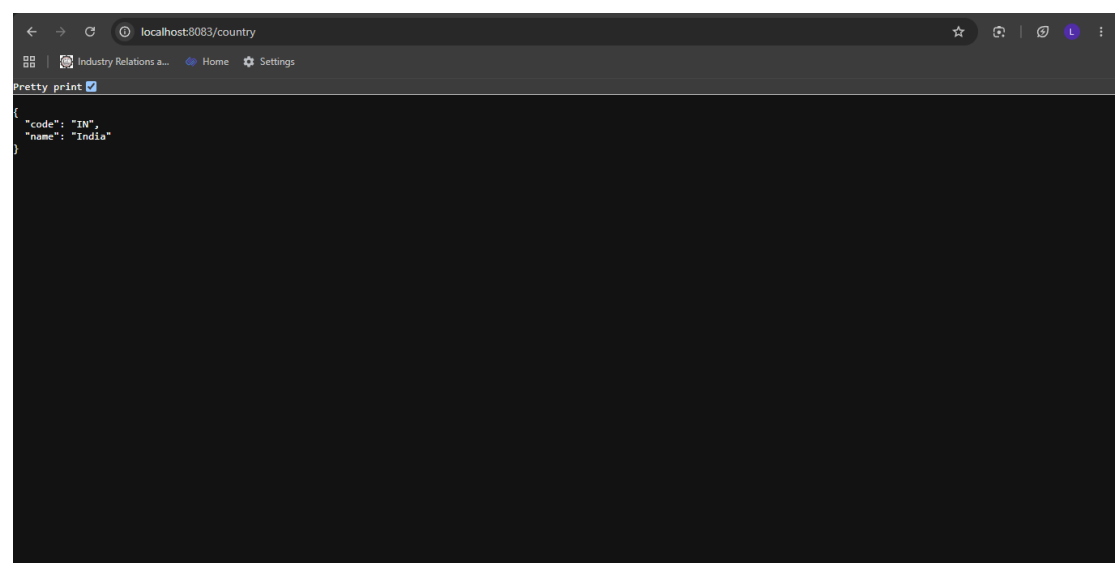
```
1 package com.cognizant.springlearn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.web.bind.annotation.RequestMapping;
8 import org.springframework.web.bind.annotation.RestController;
9
10 import com.cognizant.springlearn.model.Country;
11
12 @RestController
13 public class CountryController {
14
15     private static final Logger LOGGER =
16         LoggerFactory.getLogger(CountryController.class);
17
18     @RequestMapping("/country")
19     public Country getCountryIndia() {
20
21         LOGGER.info("START");
22
23         ApplicationContext context =
24             new ClassPathXmlApplicationContext("country.xml");
25
26         Country country =
27             context.getBean("country", Country.class);
28
29         LOGGER.info("END");
30
31         return country;
32     }
33 }
```

Output:



The screenshot shows the Eclipse IDE with the SpringLearnCountryApplication.java file open. The console output displays the following information:

```
2026-07-03T12:04:50.349+05:30 INFO 20740 --- [spring-learn-country] restartedMain c.c.SpringLearnCountryApplication : Starting SpringLearnCountryApplication using Java
2026-07-03T12:04:50.357+05:30 INFO 20740 --- [spring-learn-country] restartedMain c.c.SpringLearnCountryApplication : No active profile set, falling back to 1 default
2026-07-03T12:04:50.497+05:30 INFO 20740 --- [spring-learn-country] restartedMain e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring
2026-07-03T12:04:50.498+05:30 INFO 20740 --- [spring-learn-country] restartedMain e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider set
2026-07-03T12:04:51.090+05:30 INFO 20740 --- [spring-learn-country] restartedMain o.o.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-03T12:04:52.552+05:30 INFO 20740 --- [spring-learn-country] restartedMain o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-03T12:04:52.552+05:30 INFO 20740 --- [spring-learn-country] restartedMain o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22
2026-07-03T12:04:52.679+05:30 INFO 20740 --- [spring-learn-country] restartedMain b.w.c.s.webApplicationContextInitializer : Root WebApplicationContext: initialization comp
2026-07-03T12:04:54.199+05:30 INFO 20740 --- [spring-learn-country] restartedMain o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context
2026-07-03T12:07:49.312+05:30 INFO 20740 --- [spring-learn-country] restartedMain o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-03T12:07:49.334+05:30 INFO 20740 --- [spring-learn-country] restartedMain o.s.web.servlet.DispatcherServlet : Completed initialization in 20 ms
2026-07-03T12:07:49.545+05:30 INFO 20740 --- [spring-learn-country] restartedMain c.c.s.controller.CountryController : START
2026-07-03T12:07:51.167+05:30 INFO 20740 --- [spring-learn-country] restartedMain c.c.s.controller.CountryController : END
2026-07-03T12:08:32.121+05:30 INFO 20740 --- [spring-learn-country] restartedMain c.c.s.controller.CountryController : START
2026-07-03T12:08:32.132+05:30 INFO 20740 --- [spring-learn-country] restartedMain c.c.s.controller.CountryController : END
```



The screenshot shows a web browser with the URL `localhost:8083/country`. The response is displayed in the Pretty print view, showing the following JSON object:

```
{
  "code": "IN",
  "name": "India"
}
```

REST - Get country based on country code:

SpringLearnCountryCodeApplication.java:

```
1 package com.cognizant;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class SpringLearnCountrycodeApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(SpringLearnCountrycodeApplication.class, args);
11     }
12
13 }
14
```

Country.xml:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="countryList" class="java.util.ArrayList">
10         <constructor-arg>
11             <list>
12
13                 <bean class="com.cognizant.model.Country">
14                     <property name="code" value="IN"/>
15                     <property name="name" value="India"/>
16                 </bean>
17
18                 <bean class="com.cognizant.model.Country">
19                     <property name="code" value="US"/>
20                     <property name="name" value="United States"/>
21                 </bean>
22
23                 <bean class="com.cognizant.model.Country">
24                     <property name="code" value="JP"/>
25                     <property name="name" value="Japan"/>
26                 </bean>
27             </list>
28         </constructor-arg>
29     </bean>
30
31     Bind to grammar/schema...
32 </beans>
```

SpringLearnCountryCode/pom.xml:

```
1 https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
2 <?xml version="1.0" encoding="UTF-8"?>
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
5     <modelVersion>4.0.0</modelVersion>
6     <parent>
7         <groupId>org.springframework.boot</groupId>
8         <artifactId>spring-boot-starter-parent</artifactId>
9         <version>4.1.0</version>
10        <relativePath><!-- lookup parent from repository -->
11    </parent>
12    <groupId>com.cognizant</groupId>
13    <artifactId>spring-learn-countrycode</artifactId>
14    <version>0.0.1-SNAPSHOT</version>
15    <name>spring-learn-countrycode</name>
16    <description>Spring Core Hands-on 2</description>
17    <url/>
18    <licenses>
19        <license/>
20    </licenses>
21    <developers>
22        <developer/>
23    </developers>
24    <scm>
25        <connection/>
26        <developerConnection/>
27        <tag/>
28        <url/>
29    </scm>
30    <properties>
31        <java.version>17</java.version>
32    </properties>
33    <dependencies>
34        <dependency>
35            <groupId>org.springframework.boot</groupId>
36            <artifactId>spring-boot-starter-webmvc</artifactId>
37        </dependency>
38    </dependencies>
39 </project>
```

```
application.... Country.java spring-learn... country.xml CountryCont... SpringLearnC...
25 <developerConnection/>
26 <tag/>
27 <url/>
28 </scm>
29 <properties>
30 <java.version>17</java.version>
31 </properties>
32 <dependencies>
33 <dependency>
34 <groupId>org.springframework.boot</groupId>
35 <artifactId>spring-boot-starter-webmvc</artifactId>
36 </dependency>
37
38 <dependency>
39 <groupId>org.springframework.boot</groupId>
40 <artifactId>spring-boot-devtools</artifactId>
41 <scope>runtime</scope>
42 <optional>true</optional>
43 </dependency>
44 <dependency>
45 <groupId>org.springframework.boot</groupId>
46 <artifactId>spring-boot-starter-webmvc-test</artifactId>
47 <scope>test</scope>
48 </dependency>
49 </dependencies>
50
51 <build>
52 <plugins>
53 <plugin>
54 <groupId>org.springframework.boot</groupId>
55 <artifactId>spring-boot-maven-plugin</artifactId>
56 </plugin>
57 </plugins>
58 </build>
59 </project>
60
61
Overview Dependencies Dependency Hierarchy Effective POM pom.xml
```

Application.properties:

```
application.... Country.java spring-learn... country.xml CountryCont... SpringLearnC...
1 spring.application.name=spring-learn-countrycode
2 server.port=8084
```

CountryController.java:

```
application.... Country.java spring-learn... country.xml CountryCont... SpringLearnC...
1 package com.cognizant.controller;
2
3 import org.springframework.beans.factory.annotation.Autowired;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.PathVariable;
6 import org.springframework.web.bind.annotation.RestController;
7
8 import com.cognizant.model.Country;
9 import com.cognizant.service.CountryService;
10
11 @RestController
12 public class CountryController {
13
14     @Autowired
15     private CountryService countryService;
16
17     @GetMapping("/countries/{code}")
18     public Country getCountry(@PathVariable String code) {
19
20         return countryService.getCountry(code);
21     }
22 }
23
```

Country.java:

```
application.... Country.java spring-learn... country.xml CountryCont... SpringLearnC...
1 package com.cognizant.model;
2
3 public class Country {
4
5     private String code;
6     private String name;
7
8     public Country() {
9     }
10
11     public Country(String code, String name) {
12         this.code = code;
13         this.name = name;
14     }
15
16     public String getCode() {
17         return code;
18     }
19
20     public void setCode(String code) {
21         this.code = code;
22     }
23
24     public String getName() {
25         return name;
26     }
27
28     public void setName(String name) {
29         this.name = name;
30     }
31 }
```

CountryService.java:

```
1 package com.cognizant.service;
2
3 import java.util.ArrayList;
4
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6 import org.springframework.stereotype.Service;
7
8 import com.cognizant.model.Country;
9
10 @Service
11 public class CountryService {
12
13     public Country getCountry(String code) {
14
15         ClassPathXmlApplicationContext context =
16             new ClassPathXmlApplicationContext("country.xml");
17
18         ArrayList<Country> countryList =
19             (ArrayList<Country>) context.getBean("countryList");
20
21         for (Country country : countryList) {
22
23             if (country.getCode().equalsIgnoreCase(code)) {
24                 return country;
25             }
26
27         }
28
29         return null;
30     }
31 }
```

Output:

The screenshot shows the Eclipse IDE interface. The Project Explorer on the left displays the project structure, including the 'src/main/java' directory with the 'CountryService.java' file. The Console window at the bottom shows the output of the application, including the Spring Boot logo and the following log messages:

```
2026-07-03T15:02:14.115+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] c.c.SpringLearnCountrycodeApplication : Starting SpringLearnCountrycodeApplication
2026-07-03T15:02:14.122+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] c.c.SpringLearnCountrycodeApplication : No active profile set, falling back to 1 de
2026-07-03T15:02:14.216+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] c.c.SpringLearnCountrycodeApplication : DevTools property defaults Activel Set 'spr
2026-07-03T15:02:14.216+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] s.DevToolsPropertyDefaultPostProcessor : For additional web related logging consider
2026-07-03T15:02:15.593+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8084 (http)
2026-07-03T15:02:15.624+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Starting service (Tomcat)
2026-07-03T15:02:15.625+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.
2026-07-03T15:02:15.711+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.s.web.WebApplicationContextInitializer : Root WebApplicationContext: Initialization
2026-07-03T15:02:16.369+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8084 (http) with con
2026-07-03T15:02:16.370+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] c.c.SpringLearnCountrycodeApplication : Started SpringLearnCountrycodeApplication i
2026-07-03T15:02:16.394+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.s.web.servlet.DispatcherServlet : Initializing Spring DispatcherServlet 'disp
2026-07-03T15:02:16.394+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-03T15:02:16.394+05:30 INFO 17620 --- [spring-learn-countrycode] [ restartedMain] o.s.web.servlet.DispatcherServlet : completed initialization in 0 ms
```

The screenshot shows a web browser with three tabs: 'localhost:8084/countries/in', 'localhost:8084/countries/us', and 'localhost:8084/countries/jp'. The active tab is 'localhost:8084/countries/us'. The browser address bar shows 'localhost:8084/countries/us'. The page content displays the following JSON response:

```
{
  "code": "US",
  "name": "United States"
}
```